

Change Of Axes

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Change of Axes

$$x = x' + \alpha$$

$$y = y' + \beta$$

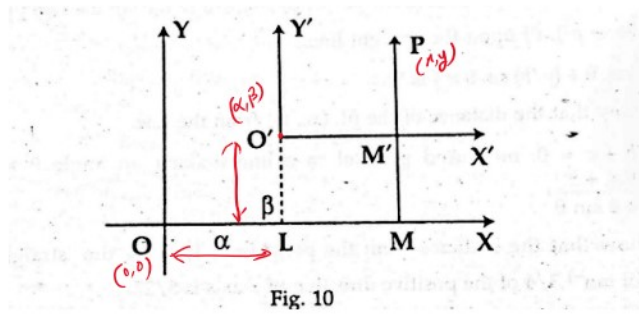


Fig. 10

Ex 1: $O(1, -2)$, $2x^2 + y^2 - 4x + 4y = 0$.

Solⁿ:

$$x = x' + 1$$

$$y = y' - 2$$

$$2(x' + 1)^2 + (y' - 2)^2 - 4(x' + 1) + 4(y' - 2) = 0$$

$$\Rightarrow 2x'^2 + 4x' + 2 + y'^2 - 4y' + 4 - 4x' - 4 + 4y' - 8 = 0$$

$$\Rightarrow 2x'^2 + y'^2 = 6$$

$$\hookrightarrow 2x'^2 + y'^2 = 6$$

Ex 2: Remove the first degree term: $3x^2 - 4y^2 - 6x - 8y - 10 = 0$

Solⁿ:

$$3x^2 - 4y^2 - 6x - 8y - 10 = 0$$

$$\hookrightarrow 3(x^2 - 2x) - 4(y^2 + 2y) - 10 = 0$$

$$\Rightarrow 3(x^2 - 2x + 1) - 4(y^2 + 2y + 1) - 10 = 0$$

$$\Rightarrow 3(x - 1)^2 - 4(y + 1)^2 - 3 + 4 - 10 = 0$$

$$\Rightarrow 3(x - 1)^2 - 4(y + 1)^2 - 9 = 0$$

$$\Rightarrow 3x'^2 - 4y'^2 - 9 = 0$$

$$\hookrightarrow 3x'^2 - 4y'^2 = 9$$

$$\left. \begin{array}{l} \text{Origin shifting to } (1, -1) \\ x = x' + 1 \\ y = y' - 1 \end{array} \right\}$$

Rotation of Axes (Origin Fixed)

$$\Delta OPR, \sin \theta = \frac{OR}{OP}$$

$$\Rightarrow OR = OP \sin \theta = x' \sin \theta$$

$$\cos \theta = \frac{OR}{OP} \Rightarrow OR = x' \cos \theta$$

$$\Delta PLO, \sin \theta = \frac{LO}{PO}$$

$$\Rightarrow LO = y' \sin \theta$$

$$\cos \theta = \frac{LO}{PO} \Rightarrow LO = y' \cos \theta$$

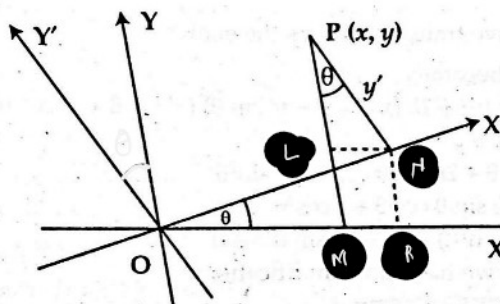
$$OM = OR - LO$$

$$\Rightarrow x = x' \cos \theta - y' \sin \theta$$

Same way, $PM = PL + LM$

$$\Rightarrow y = y' \cos \theta + x' \sin \theta$$

$$\begin{cases} x = x' \cos \theta - y' \sin \theta \\ y = y' \cos \theta + x' \sin \theta \end{cases}$$



Ex 3 $x^2 - 2xy + y^2 + 2x - 4y + 3 = 0$, rotate 45°

Soln:

$$\begin{aligned} x &= x' \cos 45^\circ - y' \sin 45^\circ = \frac{x' - y'}{\sqrt{2}} \\ y &= x' \sin 45^\circ + y' \cos 45^\circ = \frac{x' + y'}{\sqrt{2}} \end{aligned}$$

$$\therefore x^2 - 2xy + y^2 + 2x - 4y + 3 = 0$$

$$\Rightarrow \left(\frac{x' - y'}{\sqrt{2}} \right)^2 - 2 \left(\frac{x' - y'}{\sqrt{2}} \right) \left(\frac{x' + y'}{\sqrt{2}} \right) + \left(\frac{x' + y'}{\sqrt{2}} \right)^2 + 2 \frac{x' - y'}{\sqrt{2}} - 4 \frac{x' + y'}{\sqrt{2}} + 3 = 0$$

$$\Rightarrow 2y'^2 - \sqrt{2}x' - 3\sqrt{2}y' + 3 = 0$$

Ex 4: Removal of the xy -term from the expression $ax^2 + 2hxy + by^2$.

Soln:

$$\begin{cases} x = x' \cos \theta - y' \sin \theta \\ y = x' \sin \theta + y' \cos \theta \end{cases}$$

$$\Rightarrow a(x' \cos \theta - y' \sin \theta)^2 + 2h(x' \cos \theta - y' \sin \theta)(x' \sin \theta + y' \cos \theta) + b(x' \sin \theta + y' \cos \theta)^2 = 0$$

after expanding
for xy -term

$$\Rightarrow -2a \sin \theta \cos \theta + 2h(\cos^2 \theta - \sin^2 \theta) + 2b \sin \theta \cos \theta = 0$$

$$\Rightarrow (b-a) \sin 2\theta + 2h \cos 2\theta = 0$$

$$\Rightarrow (a-b) \sin 2\theta = 2h \cos 2\theta$$

$$\Rightarrow \tan 2\theta = \frac{2h}{a-b}$$

$$\Rightarrow \theta = \frac{1}{2} \tan^{-1} \left(\frac{2h}{a-b} \right)$$

$$\Rightarrow \theta = \frac{1}{2} \tan^{-1} \left(\frac{2h}{a-b} \right)$$

Some more examples:

Ex 1. By transforming to parallel axes through a properly chosen point (h, k) , prove that the equation.

$$12x^2 - 10xy + 2y^2 + 11x - 5y + 2 = 0$$

can be reduced to one containing only the terms of the 2nd degree.

Transforming to parallel axes through (h, k) we have

$$12(x' + h)^2 - 10(x' + h)(y' + k) + 2(y' + k)^2 + 11(x' + h) - 5(y' + k) + 2 = 0$$

$$\text{or, } 12x'^2 - 10x'y' + 2y'^2 + (24h - 10k + 11)x' + (-10h + 4k - 5)y' + (12h^2 - 10hk + 2k^2 + 11h - 5k + 2) = 0 \quad (i)$$

Now equate the co-efficients of x', y' to zero then

$$24h - 10k + 11 = 0 \text{ also } -10h + 4k - 5 = 0$$

$$\text{and } 12h^2 - 10hk + 2k^2 + 11h - 5k + 2 = 0$$

Solve the first two equations $h = -\frac{3}{2}, k = -\frac{5}{2}$ and it is clear that these values of h, k satisfy the equation (ii). Hence the equation. (i) becomes $12x'^2 - 10x'y' + 2y'^2 = 0$

5. The direction cosine of axes remaining the same, choose a new origin such that the new co-ordinates of the pair of points whose old co-ordinate are $(5, -13)$ and $(-3, 11)$ may be of the forms (h, k) and $(-h, -k)$. [N.U.H-1997]

Solution : Let the co-ordinate of new origin be (α, β) .

Given that the old co-ordinates of the two points are $(5, -13)$ and $(-3, 11)$ whose new co-ordinates are (h, k) and $(-h, -k)$.

$$\text{So, } (5 - \alpha, -13 - \beta) = (h, k) \text{ and}$$

$$(-3 - \alpha, 11 - \beta) = (-h, -k)$$

Equating corresponding Components,

$$5 - \alpha = h, \quad -13 - \beta = -k$$

$$-3 - \alpha = -h, \quad 11 - \beta = k$$

$$\therefore 2 - 2\alpha = 0 \quad -2 - 2\beta = 0$$

$$\therefore \alpha = 1 \quad \therefore \beta = -1$$

Thus, The co-ordinate of new origin $(1, -1)$

6. Transfer the equation $11x^2 - 4xy + 14y^2 - 58x - 44y + 126 = 0$ to new axes of X and Y whose equation are $x - 2y + 1 = 0$ and $2x + y - 8 = 0$ respectively. [N.U.H-2000; 2008]

Solution : Given that

$$11x^2 - 4xy + 14y^2 - 58x - 44y + 126 = 0 \quad \dots \quad (1)$$

Let $x - 2y + 1 = 0$ and $2x + y - 8 = 0$ be The new axes of X and Y.

New co-ordinate of (x, y) be (x_1, y_1) .

$$\text{So, } x_1 = \frac{2x + y - 8}{\sqrt{2^2 + 1^2}} = \frac{2x + y - 8}{\sqrt{5}}$$

$$\Rightarrow 2x + y - 8 = \sqrt{5}x_1 \quad \dots \quad (2)$$

$$y_1 = \frac{x - 2y + 1}{\sqrt{1^2 + 2^2}} = \frac{x - 2y + 1}{\sqrt{5}} \text{ and}$$

$$\Rightarrow x - 2y + 1 = \sqrt{5}y_1 \quad \dots \quad (3)$$

From (2) and (3)

$$4x + 2y - 16 = 2\sqrt{5}x_1$$

$$x - 2y + 1 = \sqrt{5}y_1$$

$$(+)\quad 5x - 15 = \sqrt{5}y_1 + 2\sqrt{5}x_1$$

$$5x = 15 + \sqrt{5}y_1 + 2\sqrt{5}x_1$$

$$x = \frac{2\sqrt{5}x_1 + \sqrt{5}y_1 + 15}{5}$$

$$= \frac{2x_1 + y_1}{\sqrt{5}} + 3$$

$$\text{From (2); } y = -2x + \sqrt{5}x_1 + 8$$

$$= \frac{-4x_1 - 2y_1}{\sqrt{5}} - 6 + \sqrt{5}x_1 + 8$$

$$= \frac{-4x_1 - 2y_1 + 5x_1}{\sqrt{5}} + 2$$

$$= \frac{x_1 - 2y_1}{\sqrt{5}} + 2$$

$$\therefore x = h + 3$$

$$\text{Where } h = \frac{2x_1 + y_1}{\sqrt{5}} + 3$$

$$y = k + 2 \text{ where } k = \frac{x_1 - 2y_1}{\sqrt{5}} + 2$$

Co-Ordinates—4

Putting the values of x and y in (1)

we get

$$11(h+3)^2 - 4(h+3)(k+2) + 14(k+2)^2 - 58(h+3) - 44(k+2) + 126 = 0$$

$$\Rightarrow 11(h^2 + 6h + 9) - 4(hk + 2h + 3k + 6) + 14(k^2 + 4k + 4) - 58h - 174 - 44k - 88 + 126 = 0$$

$$\Rightarrow 11h^2 - 4hk + 14k^2 - 5 = 0$$

$$\Rightarrow 11\left(\frac{2x_1 + y_1}{\sqrt{5}}\right)^2 - 4\left(\frac{2x_1 + y_1}{\sqrt{5}}\right)\left(\frac{x_1 - 2y_1}{\sqrt{5}}\right) + 14\left(\frac{x_1 - 2y_1}{\sqrt{5}}\right)^2 - 5 = 0$$

$$\Rightarrow 11(4x_1^2 + 4x_1y_1 + y_1^2) - 4(2x_1^2 - 4x_1y_1 - 2y_1^2) + 14(x_1^2 - 4x_1y_1 + 4y_1^2) - 25 = 0$$

$$= 50x_1^2 - 75y_1^2 = 25$$

$$= 2x_1^2 + 3y_1^2 = 1 \text{ which is required equation.}$$

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CHANGE OF AXES

Example 1

Transform the following equations to parallel axes through the new origin $(2, -4)$.

(i) $x^2 - 3y^2 + 4x + 6y = 0$

(iii) $3y^2 - 7x + 24y + 62 = 0$

(ii) $2x + 3y + 8 = 0$

(iv) $x^2 + y^2 - 4x + 8y - 17 = 0$.

Example 2

Transform the equation $x^2 - y^2 - 2\sqrt{2}x - 10\sqrt{2}y + 2 = 0$ when the axes are turned through $\frac{\pi}{4}$, the axes of coordinates being rectangular.

Example 3

Transform the equation $11x^2 + 24xy + 4y^2 - 20x - 40y - 5 = 0$ to rectangular axes through the point $(2, -1)$ and inclined at an angle $\tan^{-1}\left(-\frac{4}{3}\right)$.

Example 4

Transform the equation $17x^2 + 18xy - 7y^2 - 16x - 32y - 18 = 0$ to one in which there is no term involving x , y and xy .

Example 5

Transform the equation $x^2 - y^2 = a^2$ when the axes are turned through $\frac{\pi}{4}$.

Example 6

Through what angle must the axes be rotated to remove the term containing xy in $11x^2 + 4xy + 14y^2 = 5$? What is the transformed equation?

Example 7

By shifting the origin and rotating the axes of coordinates the equation $x^2 + 3xy + y^2 + 2x + 3y - 1 = 0$ is transformed into one of the form $ax^2 + by^2 = c$. Find the angle of rotation and the coordinates of the new origin.

Example 8

Find the angle through which the axes are to be turned so that the equation $\sqrt{3}x + y - 4 = 0$ may be reduced to the form $x = k$. Also, determine the value of k .

Ex 1:
Soln: ii

$$x^2 - 3y^2 + 4x + 6y = 0$$

$$\Rightarrow (x'+2)^2 - 3(y'-4)^2 + 4(x'+2) + 6(y'-4) = 0$$

$$\Rightarrow x'^2 + 4x' + 4 - 3y'^2 + 24y' - 48 + 4x' + 8 + 6y' - 24 = 0$$

$$\Rightarrow x'^2 - 3y'^2 + 8x' + 30y' - 60 = 0$$

$$\Rightarrow x'^2 - 3y'^2 + 8x' + 30y' - 60 = 0$$

$$\Rightarrow x'^2 - 3y'^2 + 8x' + 30y' - 60 = 0$$

$$\text{ii} \quad 2x + 3y + 8 = 0$$

$$\Rightarrow 2(x'+2) + 3(y'-4) + 8 = 0$$

$$\Rightarrow 2x' + 3y' = 0$$

$$\text{iii} \quad 3y^2 - 7x + 24y + 62 = 0$$

$$\Rightarrow 3(y'-4)^2 - 7(x'+2) + 24(y'-4) + 62 = 0$$

$$\Rightarrow 3y'^2 - 24y' + 48 - 7x' - 14 + 24y' - 96 + 62 = 0$$

$$(\alpha, \beta) = (2, -4)$$

$$x = x' + \alpha = x' + 2$$

$$y = y' + \beta = y' - 4$$

$$(\alpha, \beta) = (2, -4)$$

$$x = x' + 2$$

$$y = y' - 4$$

$$(\alpha, \beta) = (2, -4)$$

$$x = x' + 2$$

$$y = y' - 4$$

$$\Rightarrow 3(y-1)^2 - 7(x+2) + 24(y-1) + 12 = 0$$

$$\Rightarrow 3y^2 - 24y + 48 - 7x - 14 + 24y - 20 + 12 = 0$$

$$\Rightarrow 3y^2 - 7x - 48 = 0$$

$$(x, y) = (x', y')$$

$$x = x' + 2$$

$$y = y' - 1$$

$$\text{iv} \quad x^2 + y^2 - 4x + 8y - 17 = 0$$

$$\Rightarrow (x+2)^2 + (y-1)^2 - 4(x+2) + 8(y-1) - 17 = 0$$

$$\Rightarrow x^2 + 4x + 4 + y^2 - 2y + 1 - 4x - 8 + 8y - 32 - 17 = 0$$

$$\Rightarrow x^2 + y^2 = 32$$

$$(\alpha, \beta) = (2, -1)$$

$$x = x' + 2$$

$$y = y' - 1$$

Ex 2:

Sol:

$$x^2 - y^2 - 2\sqrt{2}x + 10\sqrt{2}y + 2 = 0$$

$$x = x' \cos \frac{\pi}{4} - y' \sin \frac{\pi}{4} = \frac{x' - y'}{\sqrt{2}}$$

$$y = x' \sin \frac{\pi}{4} + y' \cos \frac{\pi}{4} = \frac{x' + y'}{\sqrt{2}}$$

$$\Rightarrow \left(\frac{x'}{\sqrt{2}} - \frac{y'}{\sqrt{2}}\right)^2 - \left(\frac{x'}{\sqrt{2}} + \frac{y'}{\sqrt{2}}\right)^2 - 2\sqrt{2}\left(\frac{x'}{\sqrt{2}} - \frac{y'}{\sqrt{2}}\right) + 10\sqrt{2}\left(\frac{x'}{\sqrt{2}} + \frac{y'}{\sqrt{2}}\right) + 2 = 0$$

$$\Rightarrow x'^2 - 2x'y' + y'^2 - x'^2 - 2x'y' - y'^2 - 4x' + 4y' - 20x' - 20y' + 4 = 0$$

$$\Rightarrow -4x'y' - 24x' - 24y' + 4 = 0$$

$$\Rightarrow 4x'y' + 24x' + 16y' - 4 = 0$$

$$\Rightarrow x'y' + 6x' + 4y' - 1 = 0$$

Ex 3:

Sol:

$$11x^2 + 24xy + 4y^2 - 20x - 48y - 5 = 0$$

$$\Rightarrow 11(x'+2)^2 + 24(x'+2)(y'-1) + 4(y'-1)^2 - 20(x'+2) - 48(y'-1) - 5 = 0$$

$$\Rightarrow 11x'^2 + 44x' + 44 + 24(x'y' - x' + 2y' - 2) + 4y'^2 - 4y' + 4$$

$$(\alpha, \beta) = (2, -1)$$

$$\Theta = \tan^{-1}\left(-\frac{y}{x}\right)$$

$$-40(y-1) - 5 = 0$$

$$\Rightarrow 11x'^2 + 44x' + 44 + 24(xy' - x' + 2y' - 2) + 4y'^2 - 4y' + 4 - 20x' - 40 - 40y' + 40 - 5 = 0$$

$$\Rightarrow 11x'^2 + 4y'^2 + 24xy' - 5 = 0$$

$$\Rightarrow 11x'^2 + 4y'^2 + 24xy' - 5 = 0 \quad \text{--- (1)}$$

$$\text{Now, } x = x' \cos \theta - y' \sin \theta = \frac{3}{5}x' + \frac{4}{5}y'$$

$$y = x' \sin \theta + y' \cos \theta = -\frac{4}{5}x' + \frac{3}{5}y'$$

$$\tan \theta = -\frac{4}{3}$$

$$\therefore \sin \theta = -\frac{4}{5}$$

$$\cos \theta = \frac{3}{5}$$

$$\textcircled{i} \Rightarrow 11(3x' + 4y')^2 + 4(-4x' + 3y')^2 + 24(3x' + 4y')(-4x' + 3y') - 125 = 0$$

$$\Rightarrow 11(9x'^2 + 24xy' + 16y'^2) + 4(16x'^2 - 24xy' + 9y'^2) + 24(-12x'^2 + 9xy' - 16xy' + 12y'^2) - 125 = 0$$

$$\Rightarrow x'^2 - 4y'^2 + 1 = 0$$

$$\Rightarrow x^2 - 4y^2 + 1 = 0$$

For $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

origin shifting to (α, β) to eliminate terms involving x and y .

$$\alpha = \frac{fh - bg}{ab - h^2}$$

$$\beta = \frac{gh - af}{ab - h^2}$$

Ex 4:

Soln:

$$17x^2 + 18xy - 7y^2 - 16x - 32y - 18 = 0$$

$$\Rightarrow \alpha = \frac{fh - bg}{ab - h^2} = 1; \beta = \frac{gh - af}{ab - h^2} = -1$$

$$\text{Now, } 17(x+1)^2 + 18(x+1)(y-1) - 7(y-1)^2 - 16(x+1) - 32(y-1) - 18 = 0$$

$$\Rightarrow 17x'^2 - 7y'^2 - 18xy' - 10 = 0$$

$$\text{Then, } \theta = \frac{1}{2} \tan^{-1} \frac{2h}{a-b}$$

$$= \frac{1}{2} \tan^{-1} \frac{2 \cdot 9}{17-7}$$

$$= \frac{1}{2} \tan^{-1} \frac{3}{4}$$

$$\therefore 2\theta = \tan^{-1} \frac{3}{4} \Rightarrow \tan 2\theta = \frac{3}{4}$$

$$\therefore \sin 2\theta = \frac{3}{5} \quad \left| \begin{array}{l} 2\cos^2 \theta - 1 = \frac{4}{5} \Rightarrow \cos \theta = \frac{3}{\sqrt{10}} \\ 2\sin \theta \cos \theta = \frac{6}{5} \Rightarrow \sin \theta = \frac{1}{\sqrt{10}} \end{array} \right.$$

$$\cos 2\theta = \frac{4}{5}$$

$$\begin{aligned} \sin 2\theta &= \frac{3}{4} & 2\cos\theta - 1 &= \frac{1}{4} \rightarrow \cos\theta = \frac{5}{8} \\ \cos 2\theta &= \frac{1}{4} & 2\sin\theta\cos\theta &\rightarrow \sin\theta = \frac{1}{4} \end{aligned}$$

$$\begin{aligned} x &= x'\cos\theta - y'\sin\theta = \frac{3x'}{\sqrt{10}} - \frac{y'}{\sqrt{10}} \\ y &= x'\sin\theta + y'\cos\theta = \frac{x'}{\sqrt{10}} + \frac{3y'}{\sqrt{10}} \end{aligned}$$

$$17x^2 - 7y^2 + 18xy - 10 = 0$$

$$\rightarrow 17(3x' - y')^2 - 7(x' + 3y')^2 + 18(3x' - y')(x' + 3y') - 100 = 0$$

$$\Rightarrow 2x'^2 - y'^2 = 1$$

$$\rightarrow 2x^2 - y^2 = 1$$

Ex 5:

Solⁿ:

$$\left. \begin{aligned} x^2 - y^2 &= a^2 \\ \theta &= \frac{\pi}{4} \end{aligned} \right\}$$

$$\begin{aligned} x &= x'\cos\theta - y'\sin\theta = \frac{x-y}{\sqrt{2}} \\ y &= x'\sin\theta + y'\cos\theta = \frac{x+y}{\sqrt{2}} \end{aligned}$$

$$x^2 - y^2 = a^2$$

$$\Rightarrow (x-y)^2 - (x+y)^2 = 2a^2$$

$$\Rightarrow -4x'y' = 2a^2$$

$$\Rightarrow 2x'y' + a^2 = 0$$

Ex 6:

Solⁿ:

$$11x^2 + 4xy + 3y^2 = 5$$

$$\rightarrow \theta = \frac{1}{2} \tan^{-1} \frac{2 \cdot 2}{11-3}$$

$$\rightarrow \tan 2\theta = -\frac{4}{3}$$

$$\rightarrow \frac{2\tan\theta}{1-\tan^2\theta} = -\frac{4}{3}$$

$$a = 11$$

$$b = 14$$

$$h = 2$$

$$\rightarrow 6 \tan \theta = -4 + 4 \tan^2 \theta$$

$$\rightarrow 4 \tan^2 \theta - 6 \tan \theta - 4 = 0$$

$$\rightarrow \tan \theta = 2, -\frac{1}{2}$$

$$\therefore \theta = \tan^{-1}(2), \tan^{-1}\left(-\frac{1}{2}\right)$$

Ex 7:

Solⁿ:

$$x^2 + y^2 + 3xy + 2x + 3y - 1 = 0$$

$$\left. \begin{aligned} \alpha &= \frac{fh-bg}{ab-h^2} = -1 \\ \beta &= \frac{gh-af}{ab-k^2} = 0 \end{aligned} \right\} \therefore x = x', y = y'$$

$$\rightarrow (x'-1)^2 + y'^2 + 3(x'-1)y' + 2(x'-1) + 3y' - 1 = 0$$

$$\Rightarrow x'^2 - 2x' + 1 + y'^2 + 3x'y' - 3y' + 2x' - 2 + 3y' - 1 = 0$$

$$\Rightarrow x'^2 + y'^2 + 3x'y' - 2 = 0$$

$$\therefore x'^2 + y'^2 + 3x'y' - 2 = 0$$

$$\rightarrow (x'-y')^2 + (x'+y')^2 + 3(x'-y')(x'+y') - 2 = 0$$

$$\Rightarrow 2x'^2 + 2y'^2 + 3x'^2 - 3y'^2 = 4$$

$$\rightarrow 5x'^2 - y'^2 = 4$$

$$\begin{aligned} \theta &= \frac{1}{2} \tan^{-1} \frac{2b}{a-b} \\ &= \frac{1}{2} \tan^{-1} \frac{3}{1-1} \\ &= \frac{1}{2} \tan^{-1} \frac{1}{0} \\ \therefore \theta &= \frac{\pi}{4} \end{aligned}$$

Ex 8:

Solⁿ:

$$\sqrt{3}x + y = 4$$

$$\rightarrow \sqrt{3}x \cos \theta - \sqrt{3}y \sin \theta + y \cos \theta + x \sin \theta - 4 = 0$$

$$\Rightarrow x(\sqrt{3} \cos \theta + \sin \theta) - y(\sqrt{3} \sin \theta - \cos \theta) = 4$$

$$\rightarrow \cos \theta = \sqrt{3} \sin \theta$$

$$\Rightarrow \tan \theta = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \theta = \frac{\pi}{6}$$

$$\rightarrow x\left(\sqrt{3} \cdot \frac{\sqrt{3}}{2} + \frac{1}{2}\right) = 4$$

$$\Rightarrow 2x = 4$$

$$\rightarrow x = 2 \rightarrow k = 2$$

$$x = x' \cos \theta - y' \sin \theta$$

$$y = x' \sin \theta + y' \cos \theta$$